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Violence Against Human Rights Activists: Causes, Patterns, and Transformations

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1 Note:

This document and research project were conducted during the summer of 2025 in Frankfurt, Germany, as part of the Goethe Research Experience Program (GREP) at Goethe University. The project was carried out with academic supervision and institutional support, aiming to contribute to the understanding of violence against human rights defenders globally, with a particular focus on Latin America. If any questions arise regarding the content, methodology, or data sources used in this report, please feel free to reach out to the author.

2 Introduction

Violence against human rights defenders (HRDs) has become increasingly prominent across the globe, with Latin America persistently standing out as the deadliest region. These defenders, environmental activists, Indigenous leaders, journalists, and members of civil society are often targeted in settings where extractive industries intersect with fragile institutions, political instability, and deep-rooted social exclusion.

While global in scale, this is not a problem that affects all regions equally. As the image below illustrates, Latin America reports a disproportionately high number of HRD killings compared to the rest of the world. Yet despite this alarming trend, much of the existing literature and policy focus remains at the global level, often overlooking the specific dynamics that shape violence in Latin America. This study aims not only to identify broad patterns but also to highlight regional contrasts and gaps in data visibility. These visuals show that HRD violence is not evenly distributed across the globe as it can be seen in Figure 1. Latin America, despite being rich in natural resources and activism, is one of the deadliest regions for defenders as shown in Figure 2.

At the same time, countries like Russia or China show almost no recorded killings not necessarily because defenders are safer, but possibly due to censorship, repression, or lack of access to reliable information. This project aims to analyze both the visible patterns and the blind spots in global reporting.

Understanding the structural and institutional drivers behind this violence is essential—not just for advancing academic insight, but for shaping more effective international protections and local policy responses.

This report investigates the predictors and patterns associated with HRD killings using a mixed-methods statistical approach. The core analysis employs Generalized Estimating Equations (GEE) with a Negative Binomial family to model the count-based nature of the data, accounting for overdispersion and within-country correlations over time. To complement this, Random Forest machine learning models are used to detect non-linear relationships and rank variable importance through feature selection.

This work builds on foundational studies like those by Le Billon and Lujala [1] and Krain et al. [2]. Le Billon and Lujala focus on environmental and land defenders,

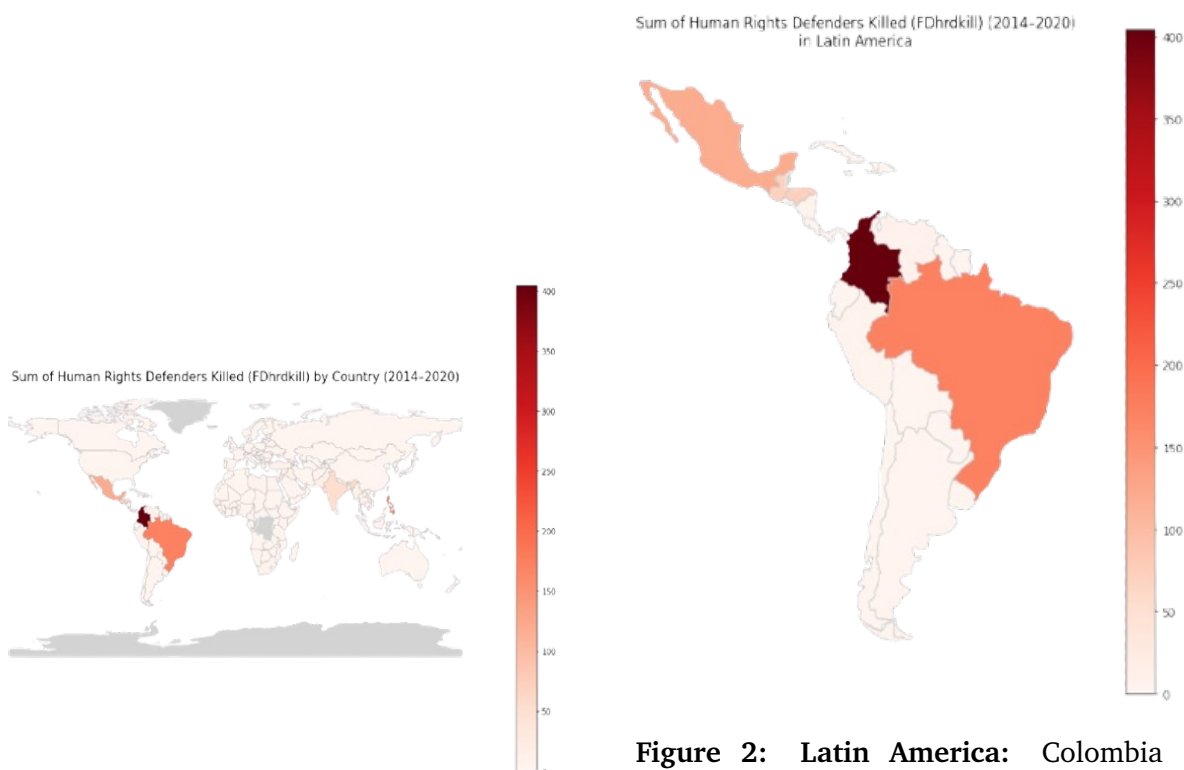


Figure 1: Global: Number of HRD killings per country (2014–2020). Latin America shows the highest levels, especially Colombia, Brazil, and Mexico.

Figure 2: Latin America: Colombia stands out with the most killings, followed by Brazil and Mexico. This shows regional patterns of violence often tied to extractive industries and political repression.

showing that killings are most likely in countries where Indigenous land claims, foreign investment, and natural resource extraction overlap particularly in semi-authoritarian regimes. Krain and colleagues emphasize state repression, institutional weakness, and limited civic space as key predictors of violence. Together, these studies suggest that HRD killings arise from a complex interplay between political repression, economic interests, and fragile governance.

By combining these theoretical frameworks with global and regional data (2014–2020), this study seeks to identify not just where defenders are most at risk, but why. Through rigorous statistical and machine learning techniques, the project maps out the drivers of lethal repression especially in Latin America and offers new insight into how policy and advocacy efforts can better respond to this growing crisis.

3 Objectives

- Identify the political, economic, institutional, and social drivers of HRD killings

- Compare global vs. Latin American models to uncover regional patterns
- Test model robustness using different statistical approaches

4 Methodology

4.1 Model Justification

Talking about the model, this research also focuses on choosing the best statistical model, given the outcome or dependent variable of interest: the number of killings of human rights defenders, which is a **count variable**.

A count variable is a type of variable that represents the number of times something happens. It's always a whole number (0, 1, 2, 3, ...) no fractions or negative numbers. This makes sense, since the number of killings must be a natural number; it cannot be a decimal, and eventually the values add up.

It is important to mention this because when the dependent variable is a count variable, a Poisson regression can be used. This type of model is used when trying to predict or explain a count the number of occurrences of something, such as:

- number of crimes in a neighborhood
- number of HRD killings

It assumes that the count variable Y follows a Poisson distribution:

$$P(Y = y) = \frac{e^{-\lambda} \lambda^y}{y!}$$

$$\log(\lambda) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

Inverse link (mean function):

$$\lambda = e^{X\beta}$$

4.2 Natural Logarithms (ln)

In this project, I applied the natural logarithm (ln) to several variables to improve how the model handles them. This is a common practice in data analysis, especially when working with variables that have large differences in scale or are highly skewed like GDP or population, which can vary widely across countries. Since the logarithm grows more slowly as values increase, it helps reduce the influence of extreme or very large numbers, making the data more balanced. It also makes trends easier to spot and results easier to interpret. For example, when both the predictor and the outcome are log-transformed, the model's coefficients can be interpreted as percentage changes, which is often more

intuitive than using raw values. Additionally, applying the logarithm helps stabilize variance across observations and can improve the model's overall performance. That's why I applied $\ln$ to several variables in this analysis.

4.3 Assumptions of Poisson Regression

- The counts are independent.
- The mean equals the variance: $\mathbb{E}(Y) = \text{Var}(Y) = \lambda$

However, this assumption is often violated. It is common for the variance to be much higher than the mean a phenomenon known as **overdispersion**. This is the case in our dataset. As a result, the Poisson model is extended to a Negative Binomial model.

4.4 Negative Binomial Regression

In the Negative Binomial regression, the count variable is modeled as:

$$\text{Mean: } \lambda = \mathbb{E}(Y)$$

$$\text{Variance: } \text{Var}(Y) = \lambda + \alpha \lambda^2$$

Unlike the Poisson model, where the variance equals the mean, here the variance can be greater than the mean due to the overdispersion parameter $\alpha > 0$.

4.5 GLM vs. GEE

As previously mentioned, prior literature employs both **Generalized Linear Models (GLMs)** and **Generalized Estimating Equations (GEEs)** to analyze count data and account for correlation structures in longitudinal or clustered settings. Both GEE and GLM with a Negative Binomial specification could be appropriate, depending on the structure of the data.

4.6 Choosing the Correlation Structure

The question that arises is: which model should we choose? Different literature and authors use different approaches. The key to making the right choice is understanding the nature of your data and the dataset you are working with. In our case, the data is **longitudinal**.

When there is correlation and the data is longitudinal, it is better to use **Generalized Estimating Equations (GEE)**. GEE offers multiple correlation structures to choose from, and the choice depends on both the nature of the data and the research objective. There are three main types of correlation matrices used in GEE, and each has implications for the results and their interpretation.

4.7 Types of Correlation Structures

1. Independence

Assumes no correlation among repeated observations within a cluster.

$$R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Appropriate when observations are assumed to be uncorrelated (rare in real-world longitudinal data).

2. Exchangeable (Compound Symmetry)

All within-cluster correlations are equal every pair of observations within a group has the same correlation ρ .

$$R = \begin{bmatrix} 1 & \rho & \rho & \rho \\ \rho & 1 & \rho & \rho \\ \rho & \rho & 1 & \rho \\ \rho & \rho & \rho & 1 \end{bmatrix}$$

Recommended when time points are not ordered or when the distance between them does not matter.

3. AR(1) (Autoregressive of Order 1)

Correlation decreases as the time gap between observations increases. Closer time points are more correlated.

$$R = \begin{bmatrix} 1 & \rho & \rho^2 & \rho^3 \\ \rho & 1 & \rho & \rho^2 \\ \rho^2 & \rho & 1 & \rho \\ \rho^3 & \rho^2 & \rho & 1 \end{bmatrix}$$

Ideal for ordered time-based data (e.g., monthly or yearly measurements).

4.8 Understanding Longitudinal and Clustered Data

Longitudinal Data (Panel or Repeated Measures):

- Track the same units (e.g., people, countries) over time
- Examples:
 - Measuring blood pressure monthly in the same patients for a year
 - Observing the GDP of each country from 2000 to 2020
- Key Feature: time and repeated observations per unit

Clustered Data:

- Grouped data (not necessarily over time)
- Examples:
 - Students within schools
 - Patients within hospitals
 - Employees within companies
- Key Feature: observations within the same group may be correlated

In our case, **both features apply**. We have countries (clusters) observed repeatedly over time (longitudinal). It is intuitive and logical to assume that observations within the same group (country) are correlated. Therefore, models like GEE or mixed models are used to account for this intra-group correlation.

4.9 Model Selection and Justification for Correlation Structure

In this case, the AR(1) correlation structure was not suitable for several models. This is likely due to the limited number of years available, which makes it difficult to reliably estimate the decay in correlation over time. Moreover, AR(1) assumes that correlations between observations decrease as the time gap increases. However, in this dataset, the correlation across time points appeared relatively constant, suggesting that an **exchangeable** structure which assumes equal correlation within clusters may be more appropriate. Additionally, not all countries report killings of human rights defenders (HRDs), which further complicates the use of AR(1).

To choose between exchangeable or independence structures, it is important to understand what each assumes:

- **Independence:** assumes there is no correlation between repeated observations within the same group.

- **Exchangeable:** assumes all observations within the same group are equally correlated.
- **Autoregressive (AR(1)):** assumes that observations closer in time are more strongly correlated than those further apart.

GEE is preferred when:

- Observations are grouped by a unit (e.g., country).
- There are repeated measurements over time (e.g., years).

In our comparison, the standard errors produced by GEE were larger than those from GLM. This indicates the presence of intra-group correlation that GLM is ignoring. Standard error reflects how much confidence we have in the estimated coefficient. A lower standard error means more precision; a higher standard error indicates more uncertainty. When comparing GEE and GLM, larger standard errors from GEE suggest that GLM may be underestimating variability and overestimating significance. This happens because GLM assumes that all observations are independent. In our case with repeated data by country this assumption is violated. As a result:

- GLM overestimates the amount of available information.
- GLM computes standard errors that are too small.
- GLM is overly optimistic about statistical significance.

In contrast, GEE produces more realistic, larger standard errors that reflect the true uncertainty. If standard errors are underestimated, we may:

- Think a variable is statistically significant (low p-value) when it actually isn't.
- Make incorrect decisions based on misleadingly precise coefficients.

If both of these problems occur underestimated standard errors from GLM and larger ones from GEE this confirms the presence of intra-cluster correlation, and using GEE is the right choice.

4.10 The GEE Estimating Equation

The **Generalized Estimating Equation (GEE)** is a method for estimating *population-averaged* effects when the data contain grouped or clustered observations. For example, repeated measurements on the same countries or individuals. The goal is to find the parameter vector β that satisfies the following estimating equation, meaning we want to find the values of β that make the score function equal to zero:

$$U(\beta) = \sum_{i=1}^K X_i^\top \Delta_i V_i^{-1} (Y_i - \mu_i) = 0$$

Where:

- K : total number of clusters or groups
- Y_i : observed outcomes for group i
- X_i : matrix of predictors for group i
- $\mu_i = g^{-1}(X_i\beta)$: expected mean response for group i , using the inverse of the link function
- $\Delta_i = \frac{\partial \mu_i}{\partial \eta_i}$: Jacobian matrix (derivative of the mean with respect to the linear predictor)
- V_i : working variance-covariance matrix for Y_i , defined as:

$$V_i = A_i^{1/2} R_i A_i^{1/2}$$

where A_i is a diagonal matrix of variances and R_i is the working correlation matrix within group i

4.11 Why the Jacobian Δ_i Matters

The Jacobian tells us how sensitive the mean μ is to changes in the linear predictor $\eta = X\beta$. It is essential for linearizing the estimating equation, allowing us to apply iterative techniques like Newton-Raphson. Without it, we cannot properly solve $U(\beta) = 0$.

4.12 Components of the Working Covariance Matrix V_i

The matrix V_i captures both the variability of outcomes and the correlation between repeated measures within a group. It is composed of:

- A_i : a diagonal matrix with the variances of each Y_{ij}
- R_i : the correlation matrix (e.g., exchangeable [**our case**], AR(1))

Example structure of A_i :

$$A_i = \begin{bmatrix} \text{Var}(Y_{i1}) & 0 & 0 & 0 \\ 0 & \text{Var}(Y_{i2}) & 0 & 0 \\ 0 & 0 & \text{Var}(Y_{i3}) & 0 \\ 0 & 0 & 0 & \text{Var}(Y_{i4}) \end{bmatrix}$$

In a Negative Binomial model:

$$\text{Var}(Y_{ij}) = \phi(\mu_{ij} + \alpha \mu_{ij}^2)$$

Then, the full working covariance is:

$$V_i = A_i^{1/2} \cdot R_i \cdot A_i^{1/2}$$

This tells the model: “Here’s how uncertain each outcome is, and how they are related within a cluster.”

4.13 Robust (Sandwich) Standard Errors

GEE doesn’t assume a full likelihood, so standard errors are adjusted using the sandwich estimator:

$$\text{Var}(\hat{\beta}) = A^{-1} B A^{-1}$$

Where A is the model-based information matrix and B captures empirical variability (“real-world noise”).

This complements the GEE model by identifying non-linear patterns and interactions that might not appear as statistically significant in traditional models but still hold predictive power.

4.14 How GEE is Calculated

The Generalized Estimating Equations (GEE) algorithm estimates population-averaged effects while accounting for within-cluster correlation. Below is a step-by-step outline of how GEE is computed:

Step 0: Start with an initial guess Pick a starting value for β , such as:

$$\beta^{(0)} = \text{vector of zeros} \quad \text{or} \quad \text{from Poisson fit}$$

Step 1: For each group (e.g., country i):

- Compute the mean:

$$\mu_i = g^{-1}(X_i \beta) \quad (\text{e.g., if log-link, } \mu_i = \exp(X_i \beta))$$

- Compute the Jacobian (derivative of the mean):

$$\Delta_i = \frac{\partial \mu_i}{\partial \eta_i} = \text{diag}(\mu_i) \quad \text{for log-link}$$

- Compute the working variance-covariance matrix:

$$V_i = A_i^{1/2} R_i A_i^{1/2}$$

where:

- A_i is a diagonal matrix of variances: $\phi V(\mu_{ij})$
- R_i is the working correlation matrix (e.g., exchangeable, AR(1), independence)

Step 2: Plug into the GEE estimating equation

$$U(\beta) = \sum_{i=1}^K X_i^\top \Delta_i V_i^{-1} (Y_i - \mu_i)$$

This is the quasi-score function — a function of β that we want to set equal to zero.

Step 3: Update β using Newton-Raphson or Fisher scoring

$$\beta^{(t+1)} = \beta^{(t)} + \left[\sum X_i^\top \Delta_i V_i^{-1} \Delta_i X_i \right]^{-1} \left[\sum X_i^\top \Delta_i V_i^{-1} (Y_i - \mu_i) \right]$$

- The matrix in brackets is the working information matrix (like a second derivative).
- The second term is the score vector (first derivative).
- This step nudges β toward a better solution.

Step 4: Repeat until convergence Repeat steps 1–3 until:

$$\|\beta^{(t+1)} - \beta^{(t)}\| < \epsilon$$

for some small threshold ϵ (e.g., 0.001).

Step 5: Compute robust (“sandwich”) standard errors Since GEE does not use full maximum likelihood, standard errors are computed using the robust (sandwich) estimator:

$$\text{Var}(\hat{\beta}) = A^{-1} B A^{-1}$$

Where:

- A : model-based information matrix
- B : empirical covariance (captures real-world noise)

Intuition: You are solving:

$$U(\beta) = 0$$

This means you're balancing what the model expects with what the data shows. GEE allows us to account for correlation within clusters (via R_i) but without assuming a full joint distribution, which makes it more flexible than maximum likelihood methods.

4.15 Control Variables

It is important to mention control variables used in the models. A **control variable** is included in the model to “adjust for” or “neutralize” its effect, even if it is not the main variable of interest.

The purpose is to prevent the results of the main explanatory variables from being “confounded” or “biased” by something important that also affects the outcome.

For example, the **logarithm of the country's population** is used as a control variable because larger countries are more likely to have more HRD killings simply due to their population size. However, the goal of the study is not to analyze the effect of population, but rather the effects of variables like democracy or corruption.

Thus, population is included to adjust for its influence and ensure the estimated coefficients of the variables of interest are not distorted.

4.16 Interaction Variables

Interaction variables help us understand how two factors might work together in a way that is different from how they behave individually. Instead of simply adding up their effects, interaction terms show whether the impact of one variable depends on the level of another. For example, when looking at youth unemployment and gender empowerment, one might ask: does unemployment affect human rights defender (HRD) killings differently in countries where women are more politically empowered?

To explore this, we multiply the two variables to create an interaction term. If this new term is significant in the model, it suggests that the relationship between unemployment and violence changes depending on the level of gender empowerment. In this way, interaction variables reveal more complex, context-sensitive, and real-world patterns that simpler models may overlook.

In this project, interaction terms were used across different models to better understand how key factors combine to influence the risk of violence against human rights defenders.

4.17 Variable Coverage

Only variables with at least 60% non-missing values were included in the models. In other words, a variable was retained if 60% or more of its observations had available

(non-missing) data. This threshold is common in this type of analysis, as it helps to have a sufficient coverage to produce reliable and interpretable estimates.

4.18 Variance Inflation Factor (VIF)

Variance Inflation Factor (VIF) is a diagnostic tool used to detect multicollinearity among predictor variables in regression models. Multicollinearity occurs when one predictor is highly correlated with others, which can inflate the variance of the estimated coefficients and make the model unstable or difficult to interpret. For a given predictor X_j , the VIF is defined as:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination obtained by regressing X_j on all the other predictors in the model. A VIF value of 1 indicates no multicollinearity, while values exceeding 5 or 10 are often used as thresholds to flag potentially problematic levels of correlation. High VIF values imply that the standard errors of the associated coefficients are inflated, reducing statistical power and potentially distorting inference.

Identifying and addressing high VIFs through variable selection, dimensionality reduction, or regularization is important for building robust models. It makes sense intuitively: if two predictors are highly correlated, they are essentially sharing the same information, which adds noise rather than clarity. Variables with high VIFs lead to unstable estimates that are sensitive to small changes in the data and make the model less reliable. In more severe cases, multicollinearity can cause convergence issues or numerical instability, particularly in logistic regression or Generalized Estimating Equations (GEE).

In this project, any variable with a VIF value greater than 10 was omitted from the analysis. Additionally, if a variable failed the VIF filter, its associated interaction terms were also excluded to avoid indirect reintroduction of multicollinearity.

4.19 QIC and the Choice of Correlation Structure

One advantage of the **exchangeable** correlation structure is that it does not require knowledge of the order or temporal distance between observations (unlike Autoregressive structures). It performs well when there is within-cluster dependence that remains relatively stable over time. Exchangeable structures tend to produce similar coefficient estimates, but with noticeably smaller standard errors, which improves statistical precision.

Furthermore, the **QIC (Quasi-likelihood under the Independence model Criterion)** was used. QIC is the equivalent of AIC for GEE models. It penalizes model complexity and rewards goodness-of-fit. A **lower QIC** value indicates a better model.

QIC was compared for both exchangeable and independence correlation structures. The model using the exchangeable structure yielded a slightly lower QIC value (e.g., 24.29 vs. 24.34), suggesting a better model fit based on this criterion. Additionally, the exchangeable model also had a greater number of statistically significant coefficients (69 out of 74), compared to the independence model (66 out of 74).

Across all models, both the QIC values and the number of significant coefficients consistently favored the exchangeable structure. These results indicate that the exchangeable structure better captures the correlation between observations within clusters, potentially resulting in more efficient estimates and an overall better model fit for the data analyzed.

However, the best choice of correlation structure also depends on the research focus. If the objective is to estimate **population-averaged effects** and obtain **robust standard errors**, the independence structure may be sufficient. But if the goal is to capture within-cluster similarity and improve estimation efficiency, the exchangeable structure is likely more appropriate.

4.20 Mathematical Definition of QIC

In Generalized Estimating Equations (GEE), model selection cannot rely on likelihood-based criteria such as AIC, since GEE does not maximize a full likelihood. Instead, the **QIC** is used. QIC balances model fit and complexity, helping to identify the best-fitting GEE model. A **lower QIC** value indicates a better-fitting model. It is particularly useful for comparing:

- Different correlation structures (e.g., independence vs. exchangeable)
- Different sets of predictor variables

QIC is mathematically defined as:

$$\text{QIC} = -2Q(\hat{\beta}) + 2 \cdot \text{trace}(V^{-1} \hat{C})$$

where:

- $Q(\hat{\beta})$ is the quasi-likelihood evaluated under the independence model
- V is the model-based covariance matrix of the parameter estimates
- $\hat{C}$ is the robust (sandwich) covariance matrix

This formulation penalizes model complexity through the trace term, similar to AIC, but is specifically adapted to the quasi-likelihood framework used in GEE. Here, $Q(\hat{\beta}, I)$ represents the quasi-likelihood under the assumption of an independence working correlation structure. As shown by [3], quasi-likelihood methods resemble maximum likelihood estimation but do not require the full specification of a probability density function. Instead, they only require the correct specification of the first two moments, i.e., the mean and variance, which is sufficient for consistent estimation.

4.21 Machine Learning

Random Forest is a non-parametric ensemble learning method that effectively handles both classification and regression problems involving continuous and categorical variables. When the outcome variable is categorical, Random Forest performs classification; when it is continuous, it performs regression.

It constructs multiple decision trees by applying **bootstrapping**, which means sampling rows with replacement to create random subsets of the original dataset. Each tree is trained on a different bootstrapped sample, increasing model diversity and reducing overfitting. At each node split, the algorithm considers a random subset of predictors to decide the best way to divide the data.

The final prediction is obtained by aggregating the outputs of all trees: by *majority vote* in classification or *averaging* in regression:

$$\hat{f}_{RF}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

where B is the number of trees and $T_b(x)$ is the prediction from the b -th tree. Splitting decisions within each tree are based on measures such as **entropy** or **Gini impurity**, which quantify how “mixed” or uncertain a group of data is. The goal at each split is to **reduce impurity** that is, to create more homogeneous subgroups where most observations belong to the same class (for classification) or have similar values (for regression). The algorithm evaluates different predictors at each split and chooses the one that produces the largest reduction in impurity. This means that a variable is selected if, after using it to split the data, the resulting groups are more “pure” or organized. The variable that achieves the greatest improvement is chosen for that split.

One of Random Forest’s key strengths lies in its ability to compute **feature importance**, which helps researchers understand *which variables were used most frequently* to make splits and *how much each variable contributed to reducing prediction error*. Variables that are used more often—especially near the top of trees—and that reduce impurity the most are considered more important. Features that appear rarely or only at deeper levels of the trees are typically less important.

This provides valuable insight into the **predictive contribution of each variable**, even in the presence of *nonlinear relationships*, *complex interactions*, or *missing data*. As a complement to **Generalized Estimating Equations (GEE)** which provide population-averaged coefficients under specific correlation structures, Random Forest enhances the analysis by uncovering hidden patterns, nonlinearities, and influential predictors. Together, GEE and Random Forest offer a comprehensive strategy: GEE for statistical inference and hypothesis testing, and Random Forest for pattern recognition, robustness, and predictive accuracy.

So, what do the variables selected by Random Forest represent? They represent the features that most effectively divide the data into purer, more predictable groups.

These variables are considered to have the highest predictive power, as they consistently contribute to reducing uncertainty (impurity) in the model's decisions.

4.22 Python Implementation and Workflow Explanation

In this project, I used Python to process and analyze my dataset after merging and cleaning it initially in Stata. The models were implemented and tested using libraries such as pandas, numpy, and statsmodels. Below, I explain the main workflow and the logic behind the code.

4.22.1 Libraries and Setup

We begin by importing key libraries:

```
import pandas as pd
import numpy as np
import statsmodels.api as sm
from statsmodels.genmod.generalized_estimating_equations import GEE
from statsmodels.genmod.families import NegativeBinomial
from statsmodels.genmod.cov_struct import Exchangeable, Independence
```

- pandas, numpy — for data manipulation and numerical operations. Pandas and numpy help handle dataframes and numerical operations.
- statsmodels — for GEE and statistical models. is used for running the GEE models with a Negative Binomial family to account for overdispersion in the count data.
- sklearn — for implementing machine learning models such as Random Forest

After merging and preprocessing the datasets (like V-Dem, WGI, and others), I separated them into global and Latin American subsets for analysis between 2014 and 2020. The initial Google Colab notebook includes all merging operations and preliminary notes. In the second Colab, I:

1. Converted categorical and year columns to proper types:

```
df\_vdem\["countrycode"] = df\_vdem\["countrycode"].astype(str)
df\_vdem\["year"] = pd.to\_numeric(df\_vdem\["year"],
errors="coerce").astype("Int64")
```

2. Centered all numerical variables (excluding dummies).
3. Created interaction variables based on domain knowledge and initial findings.

4. Built the final list of variables and filtered the dataset based on years and regions.
5. Ran VIF checks and removed variables with $VIF > 10$, along with their associated interaction terms.

4.22.2 Aligning Data and Filtering

Next, we align X and y by index, combine them in one dataframe, meaning with the filtered variables, I implemented the GEE models using the following structure.

```
df_model = pd.concat([X, y], axis=1).fillna(df_model.mean())
X = df_model[X.columns]
y = df_model["FDhrdkill"]
```

4.22.3 Defining Group Structure

We prepare the grouping and time structure for GEE:

```
groups = df_vdem.loc[X.index, "countrycode"].astype(str).values
time = df_vdem.loc[X.index, "year"].values
```

- `groups` helps account for clustering (e.g., country-level repeated measures).
- `time` can be used if the correlation structure is temporal.

4.22.4 Running the GEE Models and Comparing

We fit two GEE models using different correlation structures:

```
model2 = GEE(y, X, groups=groups, time=time,
             cov_struct=Independence(), family=NegativeBinomial())
result2 = model2.fit()
print(result2.summary())
```

```
model3 = GEE(y, X, groups=groups, time=time,
             cov_struct=Exchangeable(), family=NegativeBinomial())
result3 = model3.fit()
print(result3.summary())
```

- `Independence()` assumes no correlation between observations in a group.
- `Exchangeable()` assumes equal correlation within each group.
- `NegativeBinomial()` accounts for overdispersion in count data like HRD killings.

This approach was repeated on both the global dataset and the Latin American subset. VIF values varied slightly between the two datasets depending on variable presence.

4.22.5 Random Forest Comparison (Machine Learning)

As a complementary method, I also used Random Forest regression to evaluate model accuracy and feature importance. The relevant code snippet was:

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
from math import sqrt
```

- This helps evaluate prediction performance and compare it to GEE.
- RMSE and R^2 give a sense of error and explained variance.

This model helped validate the regression findings and explore predictive power. Test size and number of trees (`n_estimators`) were kept at standard defaults for reproducibility, but could be tuned in future research.

4.22.6 Latin America Dataset

Finally, the same process was repeated using a Latin American subset of the data. However, because the variables included were slightly different, the VIF values and interaction terms changed accordingly. This highlighted the importance of carefully checking multicollinearity and contextual differences when modeling regional vs. global data.

Note: Final datasets were uploaded to Google Drive to simplify workflow and allow easy access for model building across notebooks.

4.22.7 Access to Code and Files

The merged dataset and the initial processing steps are available in PDF format at the following link: *Merge File (PDF)*

The corresponding Google Colab notebook for the initial merge and setup can be accessed here: *Colab Notebook - Merge and Setup*

The final model's description and results are available as a PDF at the following link: *Final Model (PDF)*

The Google Colab notebook used for running the final model is available here: *Colab Notebook - Final Model*

5 Poster and Additional Material

For this project and as part of the GREP program, a research poster was developed and is available at the following link: **Poster Link**. As of now, the poster is stored on Google Drive. If any questions arise, please feel free to reach out to the authors.

It is worth noting that the methodology and preliminary results presented in the poster reflect the state of the project at that time. These can be complemented by the more recent findings discussed in this report. In comparison, this report is more comprehensive and delves deeper into each component of the research process.

6 Data Sources

This study draws on a range of high-quality and well-established datasets compiled by international institutions, non-governmental organizations, and collaborative research initiatives. The key data sources are as follows:

6.1 V-Dem (Varieties of Democracy Project)

Institutional and governance-related indicators are drawn primarily from the V-Dem dataset, one of the most comprehensive resources on global democratic practices. V-Dem relies on an expert-coded methodology to capture latent political and civil liberties constructs that are not directly observable through legal texts or formal procedures alone.

Expert-based measurement: V-Dem enlists over 4,000 country experts from across the world to provide assessments for specific country-year observations. Typically, five experts are consulted per variable per country-year, ensuring coverage and comparative reliability.

Latent variable modeling: To aggregate expert judgments and account for potential bias or inconsistency, V-Dem employs Bayesian measurement models. These models estimate both the latent trait (e.g., academic freedom) and expert-specific characteristics such as scale perception and coding reliability. The output includes both point estimates and credible intervals, reflecting measurement uncertainty.

Interpretation of scores: While latent scores range from approximately -5 to 5 , V-Dem also provides transformed scales (typically $0-4$) and ordinal versions of the indicators to facilitate interpretation. These variables are particularly useful in assessing civil liberties, rule of law, judicial independence, press freedom, and repression dynamics.

6.2 HRD Killings Data (HRD Memorial and Front Line Defenders)

The primary dependent variable Killings of Human Rights Defenders (HRDs) is sourced from the HRD Memorial, a collaborative initiative led by Front Line Defenders and over

14 other national and international human rights organizations. The project systematically gathers and verifies cases of HRDs killed in connection to their peaceful advocacy work.

Verification process: Each recorded case must be confirmed by at least three credible sources, including partner-verified data, open-source investigations, and media documentation. The methodology aims to maintain rigorous inclusion criteria and reduce the risk of false positives.

Challenges: While the HRD Memorial is one of the most reliable global datasets on this topic, the project acknowledges likely undercounting due to:

- State censorship and information suppression
- Killings in remote or inaccessible regions
- Fear of retaliation leading to non-reporting
- COVID-19-related constraints on data collection

Scope: The HRD Memorial includes cases across all regions and human rights sectors. It serves not only as a data repository but also as a commemorative initiative to honor the work and memory of defenders.

6.3 Global Witness

The study further incorporates contextual and case-based data from Global Witness, an NGO founded in 1993 to investigate the connections between natural resource extraction, corruption, environmental harm, and human rights abuses.

- Since 2012, Global Witness has tracked the killings, disappearances, and criminalization of land and environmental defenders.
- The organization combines undercover investigations, data-driven analysis, and first-hand testimonies to expose structural patterns of violence and impunity.

These data and narratives help contextualize the risks defenders face, particularly in resource-rich areas marked by extractive interests and weak institutions.

7 Results

7.1 Model Performance: RMSE and R^2

It's worth mentioning that across the different models, the results from the machine learning part, especially the Random Forest, provided helpful performance indicators: the **Root Mean Squared Error (RMSE)** and the **coefficient of determination (R^2)**.

RMSE gives us an idea of how far off the model's predictions are, on average, from the actual values. In general, *the smaller the RMSE, the better* because it means the model is making more accurate predictions. Mathematically, it's calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

R^2 tells us how much of the variation in the outcome variable is explained by the model. Values range from 0 to 1, and the closer R^2 is to 1, the better the model fits the data. It's calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

In our case, the Latin America models performed better overall. The R^2 values were generally *higher*, meaning the models explained more of the variation in HRD killings. At the same time, the RMSE values were *lower*, showing that predictions were more accurate on average. This suggests that the machine learning model, especially Random Forest, captured the patterns in the Latin America subset more effectively than in the global models.

7.2 Overview

Across models, several patterns emerge:

7.2.1 State Repression

- **State repression variables** (e.g., `v2clkill_c`, `v2ctort_c`, `PTS_S_c`) are the clearest predictors of HRD killings. In both global and Latin American models, higher tolerance or incidence of political killings and torture is strongly associated with higher HRD killings. This supports Krain et al.'s findings on the centrality of civil repression.

7.2.2 Judicial Compliance

- The variable `v2jucomp_c` shows a positive and significant association with HRD killings, suggesting a possible “façade of legality” where judicial systems exist but fail to protect defenders. This counterintuitive result may point to a disconnect between institutional presence and actual accountability.

7.2.3 Gender Empowerment and Unemployment

- `v2x_gender_c` has a positive and significant effect, suggesting that greater female political empowerment correlates with increased HRD killings. This may reflect a backlash effect in patriarchal or extractive settings.
- `wdi_unempne_c` (youth unemployment) is also positively associated with HRD killings, likely reflecting social frustration and exclusion.
- The interaction term `genderXunemployment` has a negative effect, implying that in contexts where women are more empowered, the harmful effects of structural unemployment are mitigated. Female empowerment may act as a protective buffer.

7.2.4 Ethnic and Social Composition

- Higher levels of Indigenous population are associated with more HRD killings, likely due to land and cultural disputes.
- `lnwdi_pop_c` (log population size) and `lnyoungmale_c` (log share of young males) are positively related to HRD killings, reflecting demographic pressure.

7.2.5 Political Institutions and Autonomy

- Greater subnational autonomy (`v2xel_regelec_c`) is associated with more HRD killings, potentially due to local elite capture or weak national oversight.
- Electoral competition (`v2xel_frefair`) does not always reduce HRD killings, and in weak institutional settings, may increase risk.

7.2.6 Polarization

- In Latin America, political polarization (`v2cacamps_c`) is negatively associated with HRD killings, suggesting that conflict may be channeled politically rather than violently. This pattern may not hold globally.

7.2.7 Freedom of Expression

- `e_v2x_freexp_altnf` is a double-edged variable. Greater freedom of expression may increase visibility and reporting of HRD killings without necessarily indicating an actual rise in violence.

7.2.8 Equality and Justice Measures

- `v2xeg_eqaccess_c` (equal access to power), `v2xeg_eqprotec_c` (equal protection of rights), and `v2xcl_acjst_c` (access to justice) all show negative and significant effects. These findings suggest that inclusive and fair societies experience fewer HRD killings.
- The interaction `justiceXeqaccess` supports this: countries with strong access to both power and justice report fewer killings.

7.2.9 Economic Factors

- Industrial employment (`wdi_empind_c`) is positively associated with HRD killings, while the industry's GDP share (`wdi_gdpind_c`) is negatively associated. This may reflect exploitation and informal labor conditions.
- `asinhfdi_c` (foreign direct investment) has a positive effect, in line with Le Billon & Lujala's arguments about external extractive pressures.

7.2.10 Conflict and Crime

- General homicides (`lnhomicides_ipo_c`) are strongly associated with HRD killings.
- `armedconflict_c` shows a negative coefficient. This may suggest that in overt conflict zones, HRD killings are harder to isolate or overshadowed by broader violence.

7.2.11 Random Forest Insights

- The Random Forest model validates many of the GEE findings and reveals complementary insights.
- Top predictors include `v2clkill_c`, `v2jucomp_c`, `lnwdi_pop_c`, and `justiceXeqaccess`.
- Random Forest better captures nonlinearities, particularly around population and institutional dynamics.
- Variables like `v2x_egaldem_c` (egalitarian democracy index) emerge as important predictors, even if not statistically significant in GEE.

7.2.12 Implications

- **Formal institutions are not enough:** The surprising effects of judicial compliance and electoral competition highlight that institutional design must be accompanied by enforcement and protection mechanisms.
- **Backlash politics are real:** Gender empowerment may provoke hostility if not matched by structural reforms. Empowerment alone does not guarantee protection.
- **Social inequality fuels violence:** Unemployment and unequal access to justice or power significantly increase the risk of HRD killings.
- **Decentralization without accountability is risky:** Subnational autonomy must be balanced with oversight, especially in resource-rich contexts.
- **Democratization must be substantive:** Clean elections and freedom of expression reduce risk only when embedded within strong protective and institutional frameworks.

8 Conclusion

This report brought together multiple models, data sources, and analytical approaches to explore the structural and political conditions that may influence the killing of human rights defenders. By combining traditional statistical methods like GEE with machine learning approaches like Random Forests, I was able to identify both consistent predictors and more subtle interaction effects. Importantly, the project included not only global models but also a dedicated focus on Latin America, acknowledging the region's particular vulnerabilities and data dynamics.

Throughout the process, steps were taken to ensure robustness such as centering variables, filtering out multicollinear predictors using VIF, and applying log-transformations to reduce skewness and improve interpretability. While the results are preliminary, they provide a valuable foundation for further research and can inform future models, especially when updated datasets or alternative structures are considered.

Overall, this project demonstrated how combining careful variable construction, statistical modeling, and reproducible Python workflows can help us better understand complex social outcomes like violence against defenders. The work remains open for replication and extension, and all relevant datasets, code, and documentation could be share upon request

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